

EXPERIMENT 9: ADVANCED EDGE DETECTION TECHNIQUES

AIM:

To perform advanced edge detection techniques using python in google colab.

MATERIALS REQUIRED:

Google colab software (python)

CODE:

Edge Detection Algorithms

OpenCV provides several built-in edge detection filters:

1. Sobel Operator: Detects gradients in the horizontal and vertical directions.
2. Laplacian Operator: Detects second-order derivatives to highlight regions of rapid intensity change.
3. Canny Edge Detector: A multi-step, optimal edge detector that is most commonly used.

Implementation

Stepwise implementation of Real-Time Edge Detection.

- Install Required Libraries

Installing [OpenCV](#) for image video processing, [NumPy](#) for numerical computation and [matplotlib](#) for plotting.

```
import cv2
```

```
import numpy as np
```

```
from matplotlib import pyplot as plt
```

- Import Modules

Importing required modules.

- cv2 to access OpenCV video and image functions
- numpy for array operations
- time for live FPS measurement

```
img = cv2.imread("/content/chest xray 2.jpeg")
```

-Setup Configuration Variables

Setting up paths for input images, output images and frame resolution.

```
IMG_PATH = "/content/BR.jpeg"
```

```
OUTPUT_PATH = "/content/BR.jpeg"
```

```
FRAME_WIDTH = 640
```

```
FRAME_HEIGHT = 480
```

CODE:

```
import cv2
```

```
import matplotlib.pyplot as plt
```

```
# Assuming img variable already holds ' Chest X-ray.jpg ' from previous execution
```

```
# If not, uncomment the line below:
```

```
# img = cv2.imread("Chest X-ray.jpg")
```

```
# Process the image using the defined function
```

```
processed_image_bgr = process_frame(img)
```

```
# OpenCV reads images in BGR format, matplotlib expects RGB
```

```
processed_image_rgb = cv2.cvtColor(processed_image_bgr, cv2.COLOR_BGR2RGB)
```

```
original_image_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
# Display the original and processed images
```

```
plt.figure(figsize=(12, 6))
```

```
plt.subplot(1, 2, 1)
```

```
plt.title("Original Image")
```

```
plt.imshow(original_image_rgb)
```

```
plt.axis('off')
```

```
plt.subplot(1, 2, 2)
```

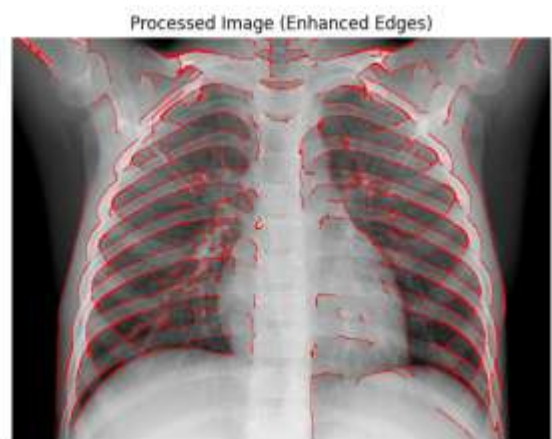
```
plt.title("Edge Detected Image")
```

```
plt.imshow(processed_image_rgb)
```

```
plt.axis('off')
```

```
plt.show()
```

OUTPUT:



RESULT:

Thus, the advanced edge detection techniques were performed by google colab.